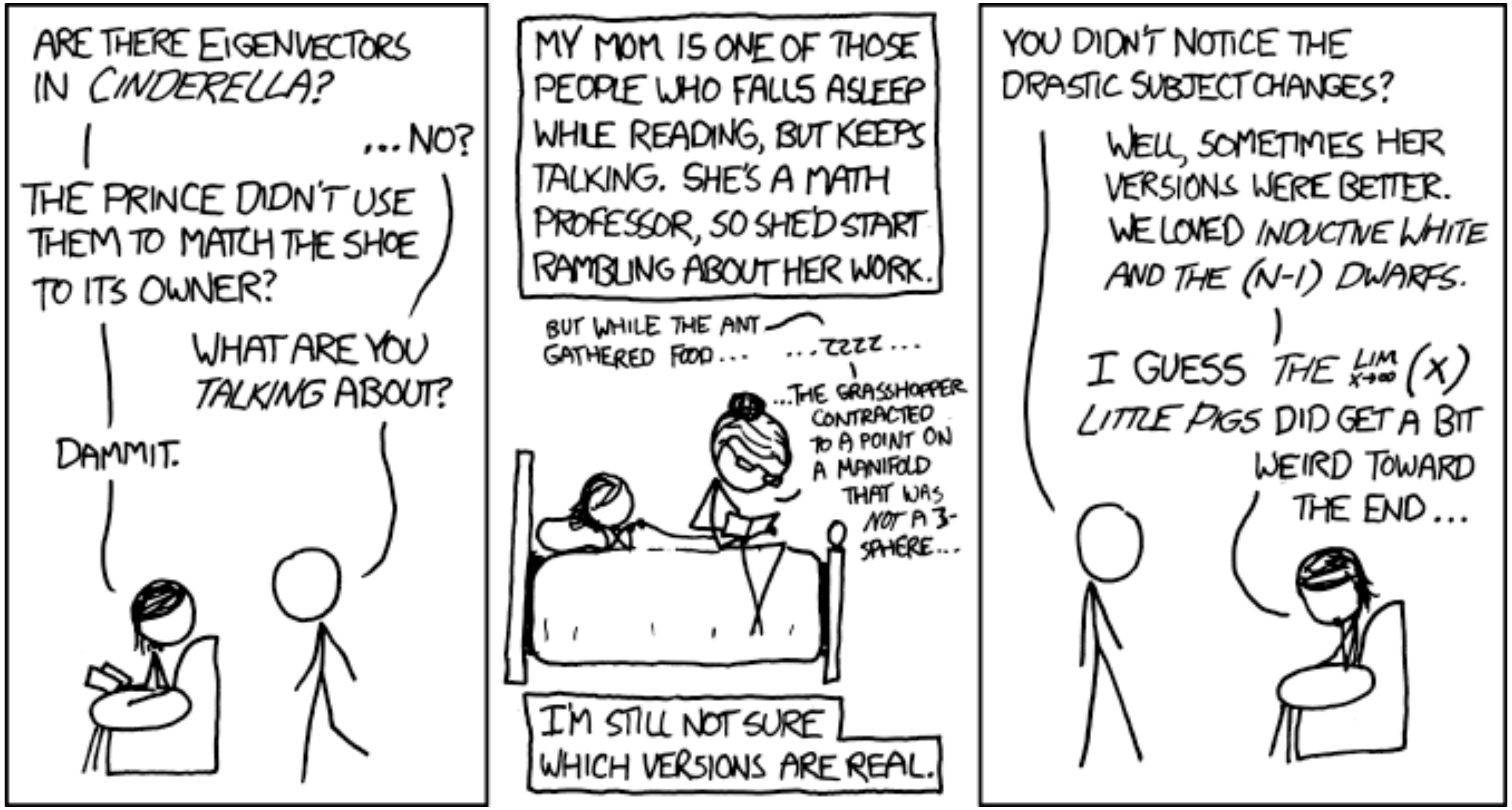

https://xkcd.com/872/

Classic Mechs EP2.1: Recap of one essential matrix math

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Classic Mechs EP2.1: Recap of one essential matrix math

Learning objectives

1) We will learn/review the basic rules of matrix multiplication, dot product.

2) We will cover two applications:

- Linear transformation & coordinate transformation

- Derive state-space from Newtonian mech to pave the way to…

solve matrix systems of equation with *eigenvalues and eigenvectors*

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```python
import numpy as np
np.dot (A,x)
```
or
```python
A @ x
```

Matrix multiplication, dot product

2x2 example:

$$M3 = M1 \times M2 = \begin{bmatrix} \boxed{a_1 \quad b_1} \\ c_1 \quad d_1 \end{bmatrix} \bullet \begin{bmatrix} \boxed{\begin{matrix} a_2 \\ c_2 \end{matrix}} & \begin{matrix} b_2 \\ d_2 \end{matrix} \end{bmatrix} = \begin{bmatrix} \boxed{a_1 \times a_2 + b_1 \times c_2} & a_1 \times b_2 + b_1 \times d_2 \\ c_1 \times a_2 + d_1 \times c_2 & c_1 \times b_2 + d_1 \times d_2 \end{bmatrix}$$

Generalization:

$$P_{i,j} = \sum_{k=1}^{n} Q_{i,k} \bullet R_{k,j}$$

$$P = \begin{bmatrix} Q_{11}R_{11} + Q_{12}R_{21} + \cdots + Q_{1n}R_{n1} & Q_{11}R_{12} + Q_{12}R_{22} + \cdots + Q_{1n}R_{n2} & \cdots & Q_{11}R_{1q} + Q_{12}R_{2q} + \cdots + Q_{1n}R_{nq} \\ Q_{21}R_{11} + Q_{22}R_{21} + \cdots + Q_{2n}R_{n1} & Q_{21}R_{12} + Q_{22}R_{22} + \cdots + Q_{2n}R_{n2} & \cdots & Q_{21}R_{1q} + Q_{22}R_{2q} + \cdots + Q_{2n}R_{nq} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{m1}R_{11} + Q_{m2}R_{21} + \cdots + Q_{mn}R_{n1} & Q_{m1}R_{12} + Q_{m2}R_{22} + \cdots + Q_{mn}R_{n2} & \cdots & Q_{m1}R_{1q} + Q_{m2}R_{2q} + \cdots + Q_{mn}R_{nq} \end{bmatrix}$$

Matrix element indexing: [i, k]

Row # **Column #**

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The dot product application: linear transformation and coordinate transformation

Unfortunately, no one can be told what the Matrix is. You have to see it for yourself.

-Morpheus

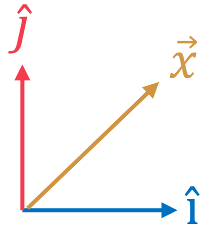
https://youtu.be/kYB8IZa5AuE?si=su9SausqkmoxiRoU

$\vec{x}$: Every single vector in space

$\overrightarrow{x'}$ Output vector $\quad$ $\vec{x}$ Input vector

$$[\ \ ] = [\qquad\quad][\ \ ]$$

A matrix



$$\begin{array}{cc} \hat{\imath} & \hat{j} \end{array}$$
$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

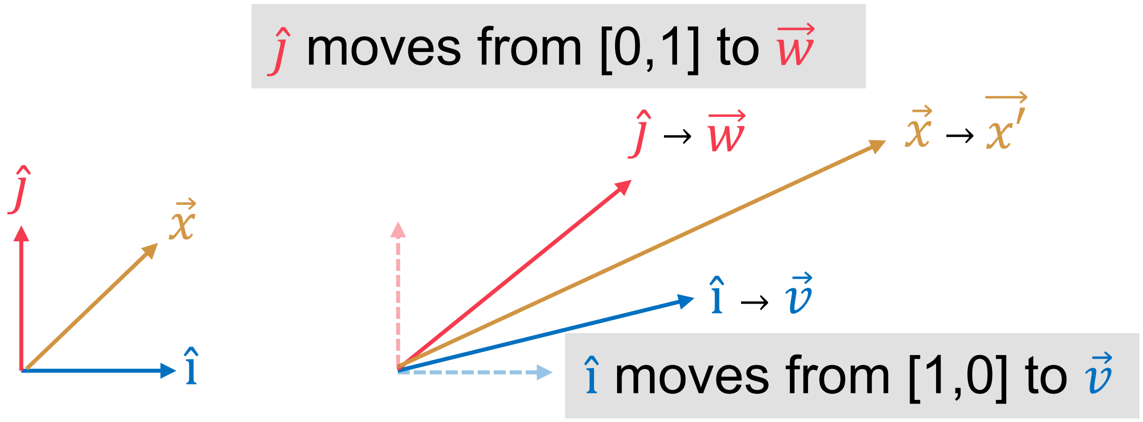
Recall: any vector can be described with basis vectors $\hat{\imath}$ and $\hat{j}$.

How do we describe a linear transformation mathematically without tracking every single point in space?

The dot product application: linear transformation and coordinate transformation

> Unfortunately, no one can be told what the Matrix is. You have to see it for yourself.
>
> -Morpheus

https://youtu.be/kYB8IZa5AuE?si=su9SausqkmoxiRoU



$$\begin{array}{cc} \hat{\imath} & \hat{\jmath} \\ \end{array} \qquad \begin{array}{cc} \vec{v} & \vec{w} \end{array}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \; & \; \\ \; & \; \end{bmatrix}$$

A matrix

$$\underline{x'} = \underline{\underline{A}}\,\underline{x}$$

The dot product application: linear transformation and coordinate transformation

A 90° Counter-Clockwise Rotation:

- $\hat{i}$ moves from $[1, 0]$ to $[0, 1]$.
- $\hat{j}$ moves from $[0, 1]$ to $[-1, 0]$.
- The resulting rotation matrix is: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

A Shear Transformation:

- $\hat{i}$ remains fixed at $[1, 0]$.
- $\hat{j}$ leans over to $[1, 1]$.
- The resulting shear matrix is: $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Also Euler rotations:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation of $\hat{\imath}$ Rotation of $\hat{\jmath}$

More example of coordinate transformation:

Sequential Euler rotations in 3D: $\mathbf{v}_{final} = \mathbf{R}_z(\psi) \cdot \mathbf{R}_x(\theta) \cdot \mathbf{R}_z(\phi) \cdot \mathbf{v}_{initial}$

$$\mathbf{R}_z(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \quad \mathbf{R}_x(\psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi & -\sin\psi \\ 0 & \sin\psi & \cos\psi \end{bmatrix}$$

Might be an interesting read: loss of degree of freedom (DOF) and gimbal lock: https://en.wikipedia.org/wiki/Gimbal_lock.

Bonus: describe kinetic energy (T) in metric tensor format

2D Cartesian Coordinates (x, y)

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$$

$$\downarrow$$

$$T = \frac{1}{2}m \begin{bmatrix} \dot{x} & \dot{y} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix}$$

metric tensor g

2D Polar Coordinates (r,θ)

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$$

$$\downarrow$$

$$T = \frac{1}{2}m \begin{bmatrix} \dot{r} & \dot{\theta} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & r^2 \end{bmatrix} \begin{bmatrix} \dot{r} \\ \dot{\theta} \end{bmatrix}$$

3D Cylindrical Coordinates(r,θ,z)

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2)$$

$$\downarrow$$

$$T = \frac{1}{2}m \begin{bmatrix} \dot{r} & \dot{\theta} & \dot{z} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{r} \\ \dot{\theta} \\ \dot{z} \end{bmatrix}$$

This is how we get metric tensor: $g = J^T J$,

where $J = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix}$

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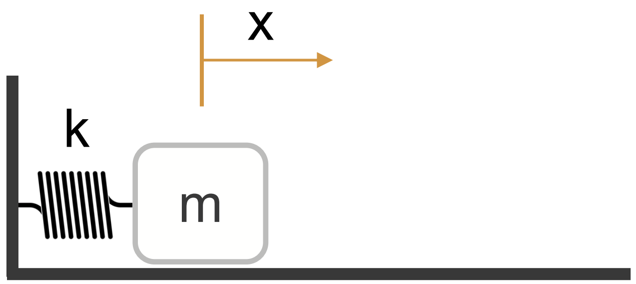
The dot product application: state-space

Sigle harmonic oscillators*:



2nd order linear DE

$$\ddot{x} = -\frac{k}{m}x$$

A system of coupled
1st order linear DEs

$$\dot{x} = v$$

$$\dot{v} = -\frac{k}{m}x$$

$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$

WHY?

https://youtube.com/shorts/AqME_OqamWQ

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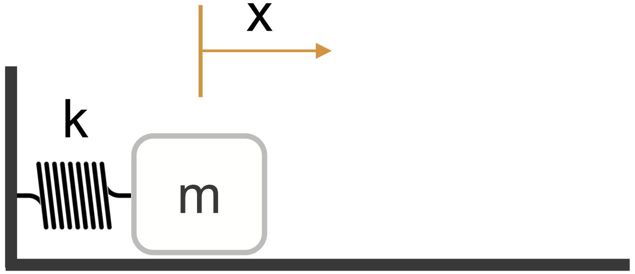
*Assuming: no friction, no air resistance

The dot product application: state-space

Sigle harmonic oscillators:

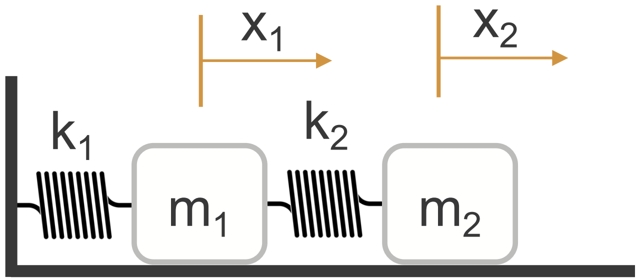


$$\dot{x} = v$$

$$\dot{v} = -\frac{k}{m}x$$

$$\begin{bmatrix}\dot{x}\\ \dot{v}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{k}{m} & 0\end{bmatrix}\begin{bmatrix}x\\ v\end{bmatrix}$$

Double harmonic oscillators case A:

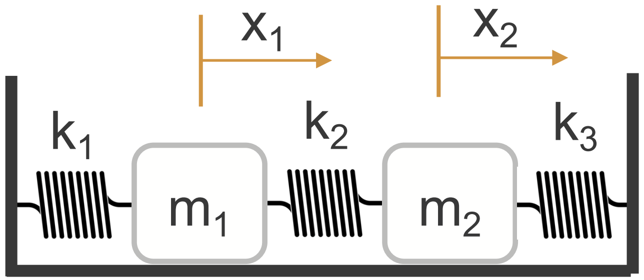


$$m_1\ddot{x}_1 = -k_1x_1 + k_2(x_2 - x_1)$$

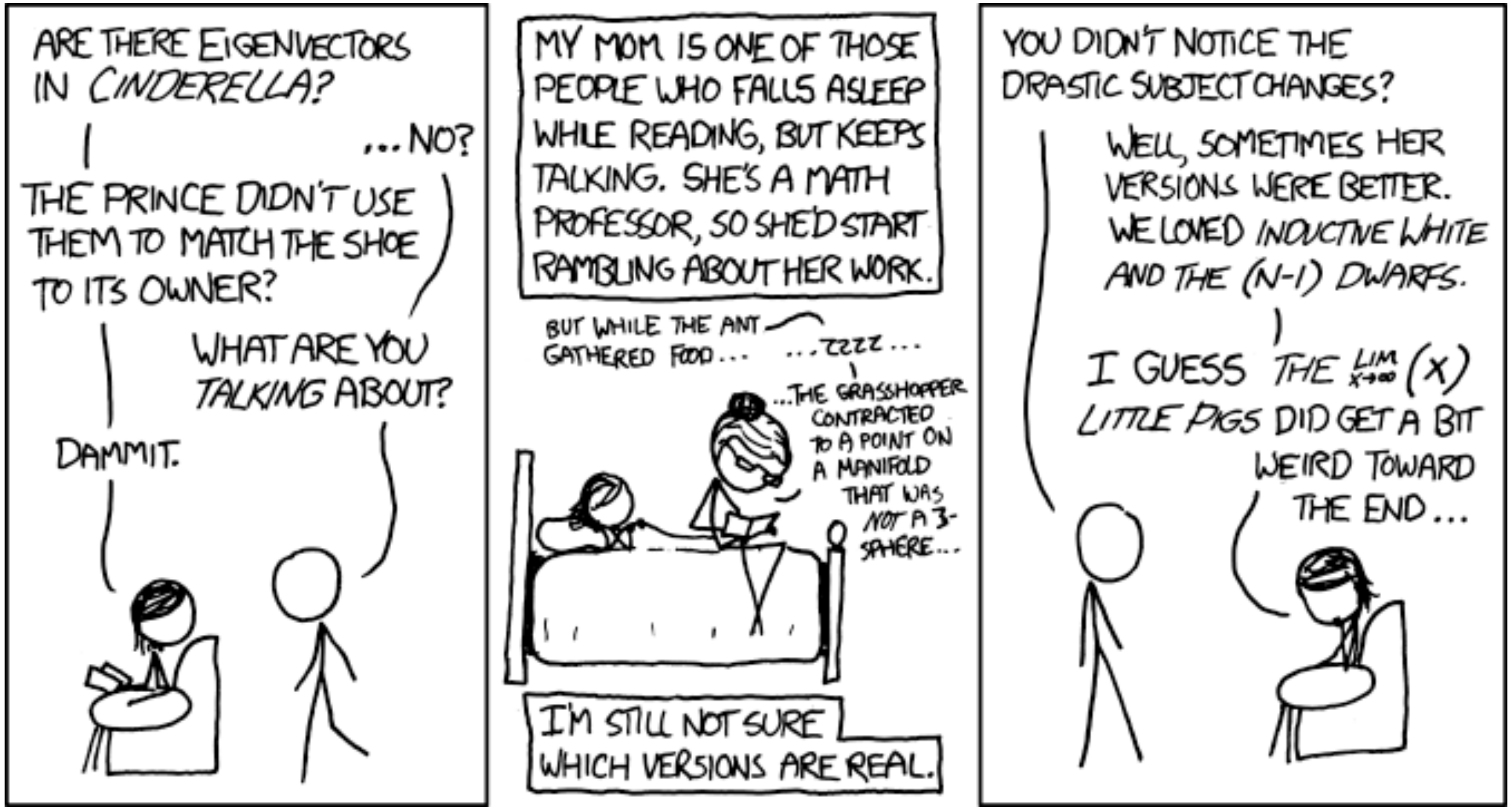
$$m_2\ddot{x}_2 = -k_2(x_2 - x_1)$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{v}_1\\ \dot{x}_2\\ \dot{v}_2\end{bmatrix} = \begin{bmatrix}0 & 1 & 0 & 0\\ -\frac{k_1+k_2}{m_1} & 0 & \frac{k_2}{m_1} & 0\\ 0 & 0 & 0 & 1\\ \frac{k_2}{m_2} & 0 & -\frac{k_2}{m_2} & 0\end{bmatrix}\begin{bmatrix}x_1\\ v_1\\ x_2\\ v_2\end{bmatrix}$$

Double harmonic oscillators case B:



Assuming: no friction, no air resistance



https://xkcd.com/872/

Classic Mechs EP2.2: Recap of some essential ODE math

Coe College

Classic Mechs EP2.2: Recap of one essential ODE math

Learning objectives

1) We will learn/review Euler’s formula (proof via Taylor series?)

2) We will learn/review two ways of solving harmonic oscillator ($e^{-\lambda t}$, and $e^{\underline{\underline{A}}t}x(0)$)

3) We will then work on damped harmonic oscillator with the same approach

4) Then we try to solve some easy examples of 2nd order ODE

to pave the way to…

solve high order ODE with *eigenvalues and eigenvectors*

Classic Mechs EP2.2: Recap of one essential ODE math

Euler's formula

$$\begin{aligned} e^{ix} &= 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \frac{(ix)^6}{6!} + \frac{(ix)^7}{7!} + \frac{(ix)^8}{8!} + \cdots \\ &= 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \frac{x^6}{6!} - \frac{ix^7}{7!} + \frac{x^8}{8!} + \cdots \\ &= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right) \\ &= \cos x + i \sin x, \end{aligned}$$

https://en.wikipedia.org/wiki/Euler%27s_formula

Classic Mechs EP2.2: Recap of one essential ODE math

Two ways of solving harmonic oscillator ($e^{-\lambda t}$, and $e^{At}x(0)$), then the damped ones

$$x(t) = C_1 e^{\lambda_1 t} + C_2\, e^{\lambda_2 t}$$

$C_1$and C_2 will be decided by the initial conditions (I.C.).

Bonus discussion:

Eigenvalue λ_1 λ_2 … can directly indicate whether the sytem is stable.

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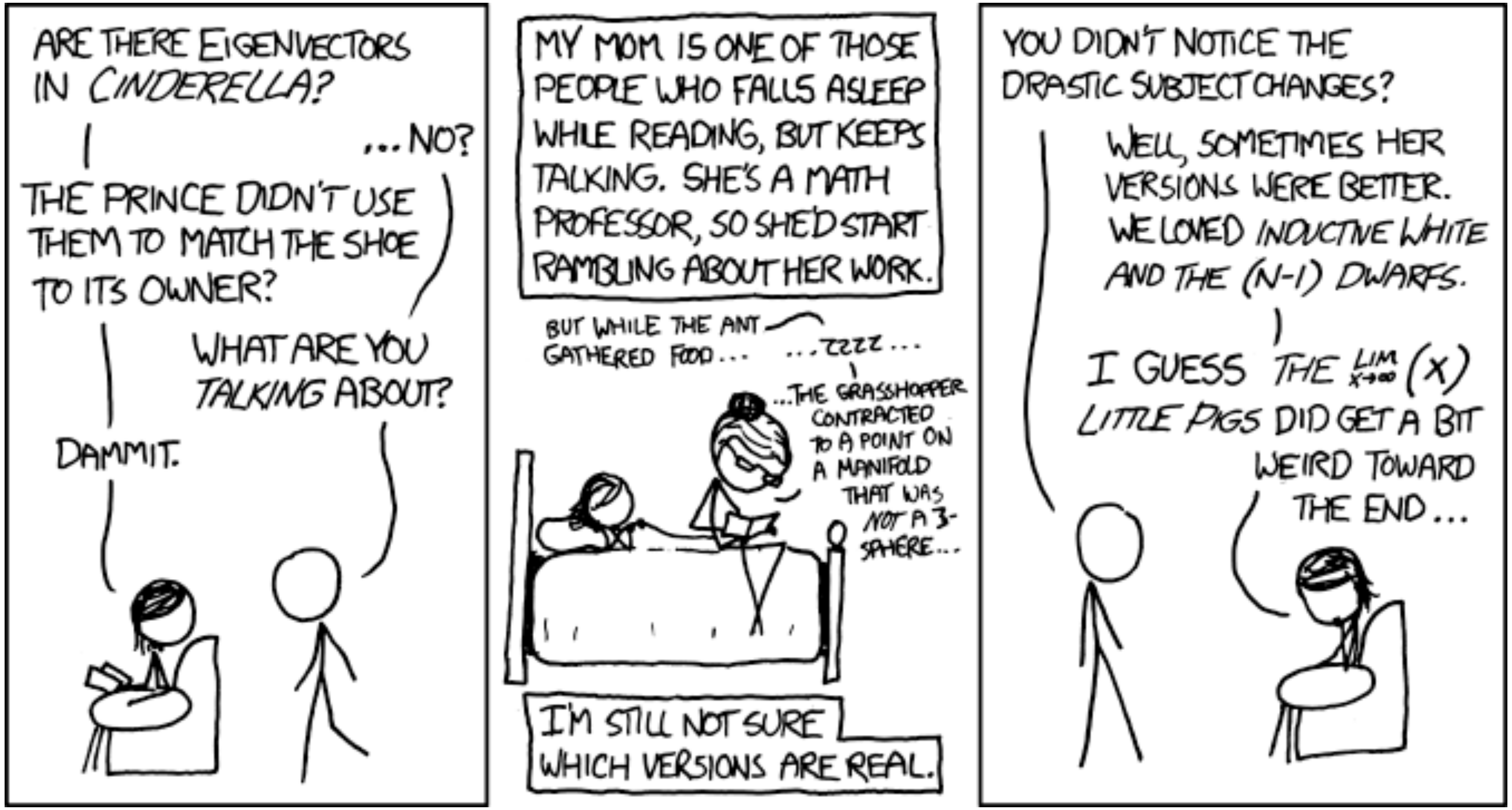
Classic Mechs EP2.2: Recap of one essential ODE math

Solve some easy examples of 2^{nd} order ODE

$$\ddot{x} + \dot{x} + 2x = 0 \qquad \mathrm{I.C.:}\ x_0 = 2, \dot{x}_0 = -3$$

$$\ddot{x} + \dot{x} - 2x = 0 \qquad \mathrm{I.C.:}\ x_0 = 3, \dot{x}_0 = 0$$

Pre-lab for next week: to use state-space matrix (i.e. coupled 1^{st} order matrix of equations) to write these two equations and compute the eigenvalues.



https://xkcd.com/872/

Classic Mechs EP2.21: two extra awesome applications about matrix

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Follow-up on equation (1.36) :

$$\ddot{x} = g(\sin\theta - \mu\cos\theta)$$

First, let's simplify the condition further: no friction, small angle assumption.

$$\ddot{x} = g\theta$$

State-space matrix?

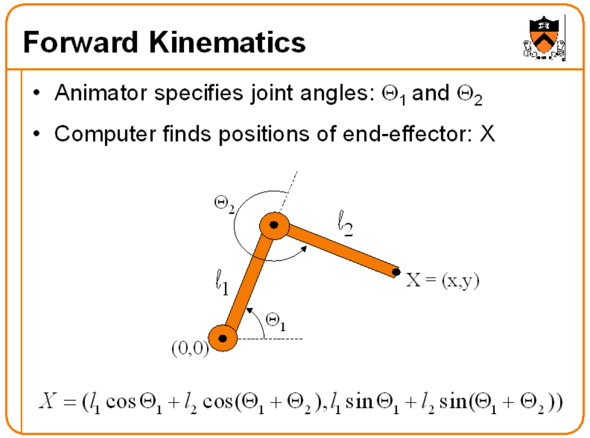
Add closed-loop control?

Follow-up on Euler’s rotation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation of $\hat{\imath}$ Rotation of $\hat{\jmath}$

$$\begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



https://www.cs.princeton.edu/courses/archive/fall99/cs426/lectures/kinematics/sld004.htm

But with DH parameters

https://en.wikipedia.org/wiki/Denavit%E2%80%93Hartenberg_parameters

Pre-lab for next week: with the help of Sequential Euler rotations in 3D (slide #7), derive the DH parameters for 3D forward kinematics, two arms and two joint.

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